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Prelab 8.1 Introduction to Machine Learning

Learning Goals

1. Identify the characteristics of an overfitted, underfitted, and well-fitted model
2. Develop, train, and refine a model that uses machine learning to detect anomalies on data batches.
3. Assess the accuracy, precision, and loss of a model in detecting anomalies on data batches.

Foundation and Background

What is Machine Learning (ML)?

Machine learning figures things out on its own. It doesn't need explicit programming to find patterns, features, or irregularities in big data sets. Compare the traditional programming paradigm to the machine learning paradigm below.

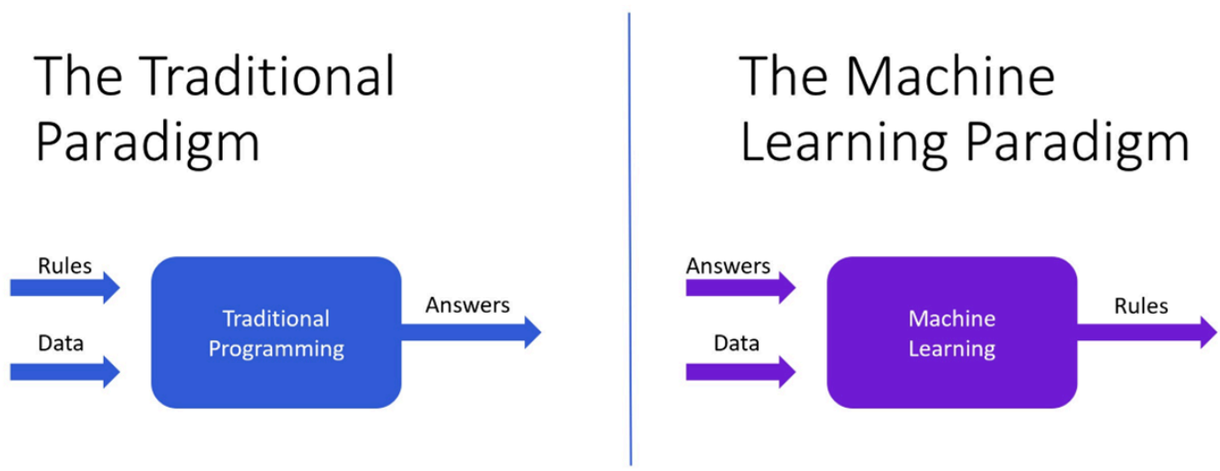

Figure 1 TP vs ML paradigm[1]

For a Machine Learning algorithm to solve a problem, there are two key features. (1) it will take a piece of **data** and (2) will take a **guess** at the **answer**. Then, it will *optimize* its **guess** until it gets the correct **answer**. The concept of *loss* will be what helps to *optimize* the **guess**.

Task 1.1

(1) Come up with an example of where Machine Learning would be more useful than Traditional Programming. (2) Explain why using Traditional Programming would not be optimal.

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(1) Example where Machine Learning is more useful: A good example is email spam detection. Instead of manually writing rules to classify emails as spam or not (e.g., checking for certain keywords like “free,” “win,” etc.), a Machine Learning model can be trained on thousands of labeled emails to automatically learn patterns. It can detect subtle features such as writing style, sender behavior, and hidden correlations that are difficult to explicitly program.

(2) Why Traditional Programming is not optimal: Traditional programming relies on explicitly defined rules, which becomes inefficient in this case because spam emails constantly evolve. Writing rules

to cover all variations would be extremely time-consuming and still incomplete. Additionally, rigid rules cannot adapt to new types of spam without continuous manual updates. In contrast, Machine Learning models can generalize from data and improve over time, making them more flexible, scalable, and accurate for such dynamic problems.

To illustrate, the below diagram shows the general machine learning process.

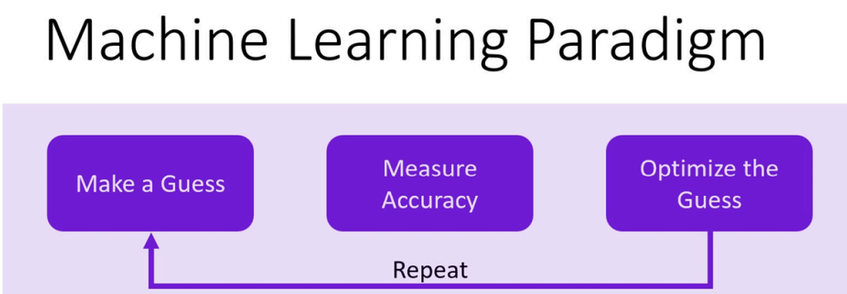


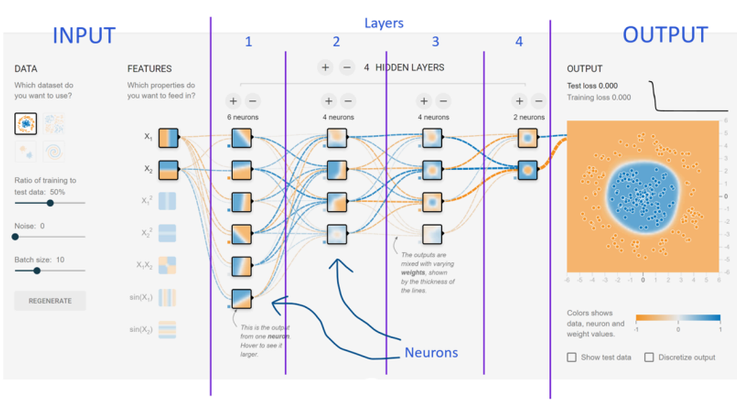
Figure 2 ML paradigm process [1]

Task 1.2: (1) With one paragraph (3-5 sentences), explain machine learning very simply. Hint, using the human brain as an analogy makes sense here.

Machine learning is like how the human brain learns from experience. When we see something many times, our brain starts to recognize patterns and make better decisions without needing step-by-step instructions each time. Similarly, a machine learning model is trained on data, and it learns patterns from that data instead of being explicitly programmed with rules. Over time, just like humans improve with practice, the model becomes better at making predictions or decisions. This ability to learn and adapt makes machine learning powerful for solving complex problems.

Some main aspects of ML

- Neurons and Layers
  - When designing a neural network, there are decisions to make on the amount of layers and the amount of neurons. Reference the picture below to see neurons & layers and how they are portrayed in a visualized ML model. If you want to play with the website used in this screenshot, go to ' https://playground.tensorflow.org/ '.



- Model
  - The product of training and data sets. When "machine learning" happens, it is usually the model making inferences on data given to it. See the figure below.

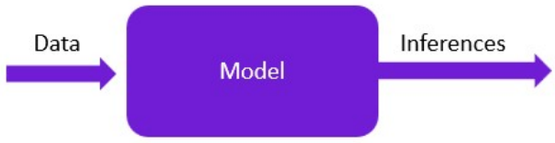
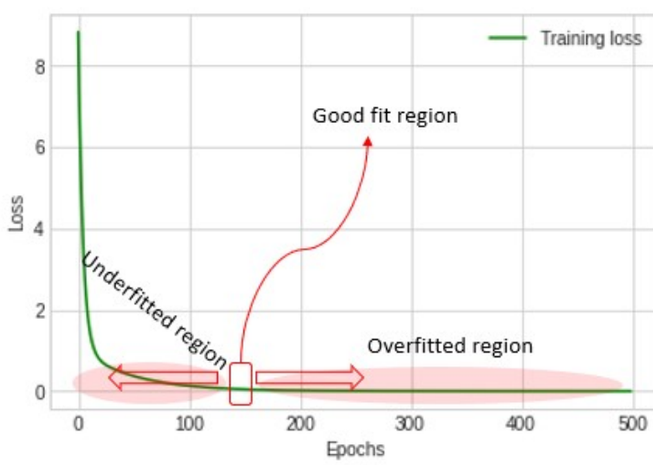


Figure 3 A "model"[1]

- Training
  - The process of determining the best "fit" for the model. After a bunch of iterations, a model will be trained for making good inferences. Seeing figure 2 above, that "Repeat" process is the training process.
- "Fit"ness
  - As an ML Engineer, you can either underfit or overfit a model. If a model is underfit, it will not make good inferences on the data used to train the model and will not make good inferences on the real-world/unknown data. An overfit model will be super good at the training data, but when in the real-world, it will not be good. Take a look at the figure below.



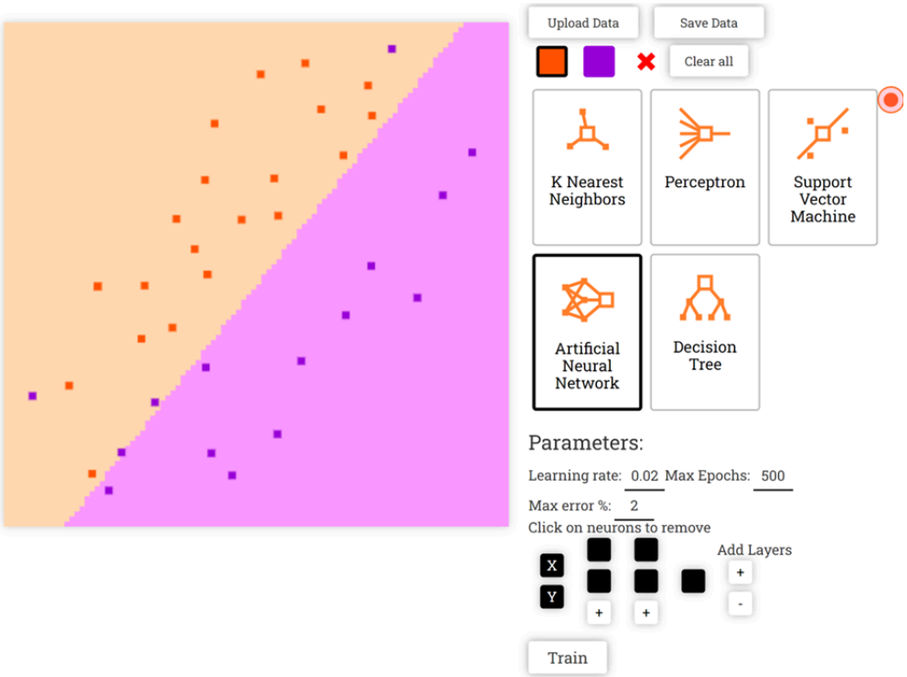
- Epochs
  - The amount of times the process in Figure 2 is repeated. When training a model, the algorithm will run through multiple iterations to minimize the loss as much as possible.
- Visualizing
  - Visualizing helps the ML Enginineer (you) determine if you have appropiately tuned your model (the parameters) according to your data set. Typically, neural networks are seen as "black boxes" -- it is hard for humans to make reliable inferences based on the machine learning code, layer decisions, neuron decisions, etc.

Exploring how neurons, layers, and algorithms change the efficiency of our chosen ML model

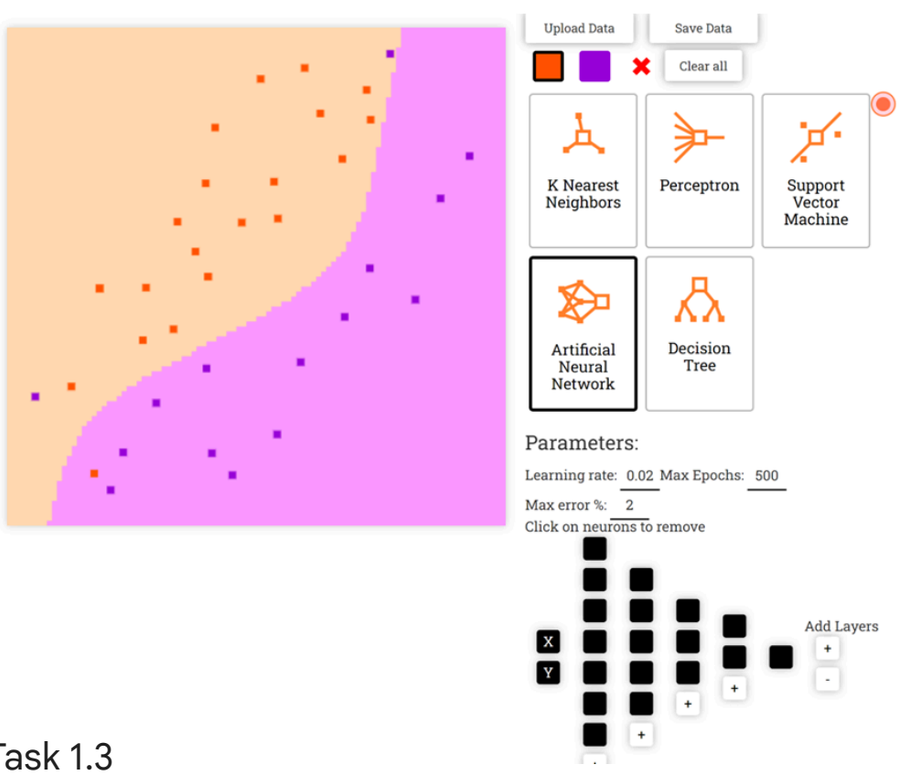
We will use the website ' https://ml-playground.com/ ' to form our intuitions and gain a better understanding of how the different neurons, layers, and machine learning algorithms effect a model.

This website allows you to click Orange and Purple dots to form data points, then choose an algoritm to train your model. Once selecting an algorithm, you'll see there are choices to let you tune your parameters. This algorithm, with parameters, is how your model will be trained. This process is visualized and simplified to help you build intuition.

The screenshot below shows a fit model. By looking at the data points, it is safe to say the algorithm should classify half orange and half purple, although there are data points, or anomalies, on either side of the opposite color. You can see in the bottom right that there are 2 layers with 2 neurons each. These parameters were chosen after trial-and-error to find the best one.



The screenshot below shows an overfit model. You see, the data is roughly split 50/50 by a diagonal line. However, the colored regions show where the model would predict a data point to be. And, in this case it would not infer correctly. It is too fit for the training data, and if we introduced new data (real-world / unknown data), it would not draw accurate inferences.



Task 1.3

There is not an example provided for the underfit model, that is your Task.
Draw data points wherever you see fit. Try to use a pattern of orange/purple that is not like the data in the screenshots we have provided.

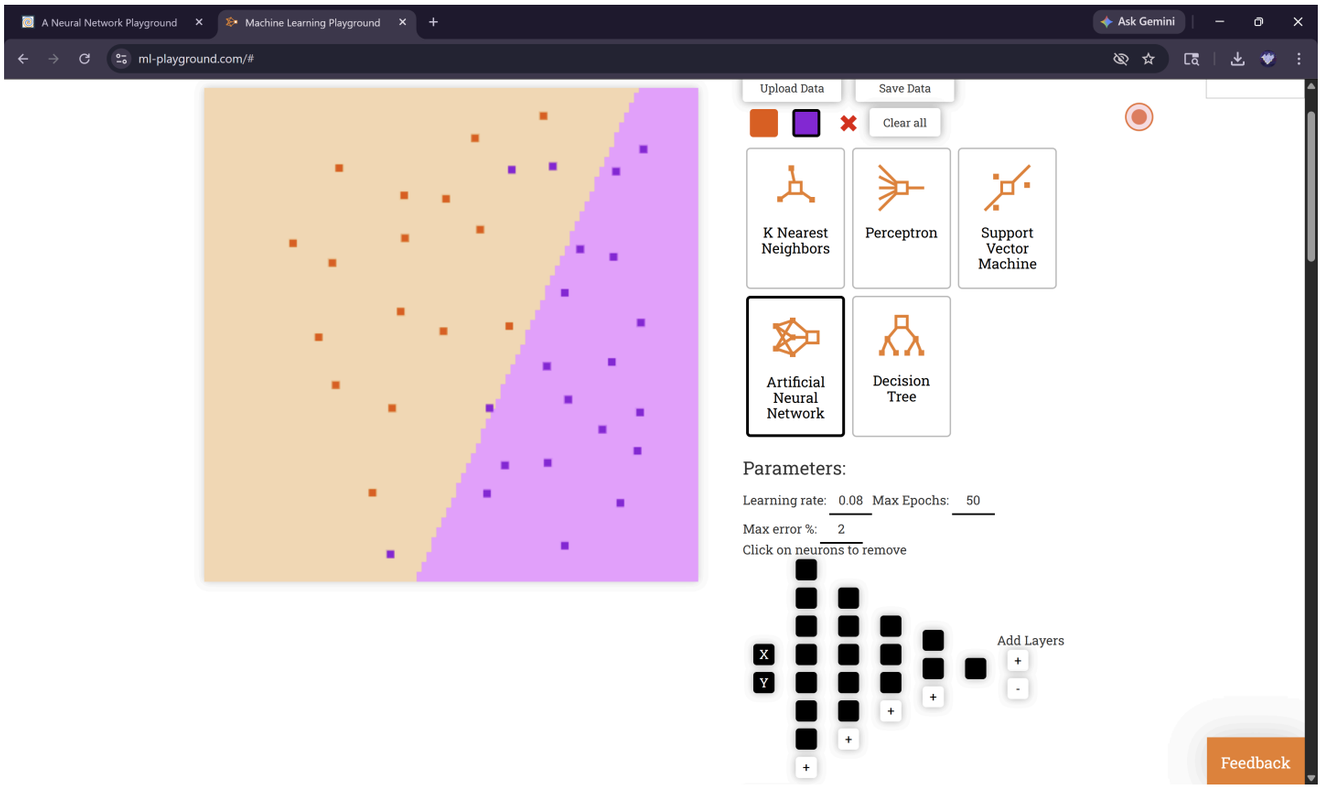
Once you have a pattern of orange/purple dots, select the **Artificial Neural Network** and mess with the parameters below. So, change the amount of neurons and layers and click "Train" to see if you think the colored regions are suitable for the data you put in. Feel free to read the information the website provides below the graph, and mess with learning rate, epochs, and error as well.

Provide a screenshot of your data with **(1)** underfit regions, **(2)** overfit regions, and **(3)** fit regions.

(4) Explain why the choices you made impacted your model to be fit or not fit.

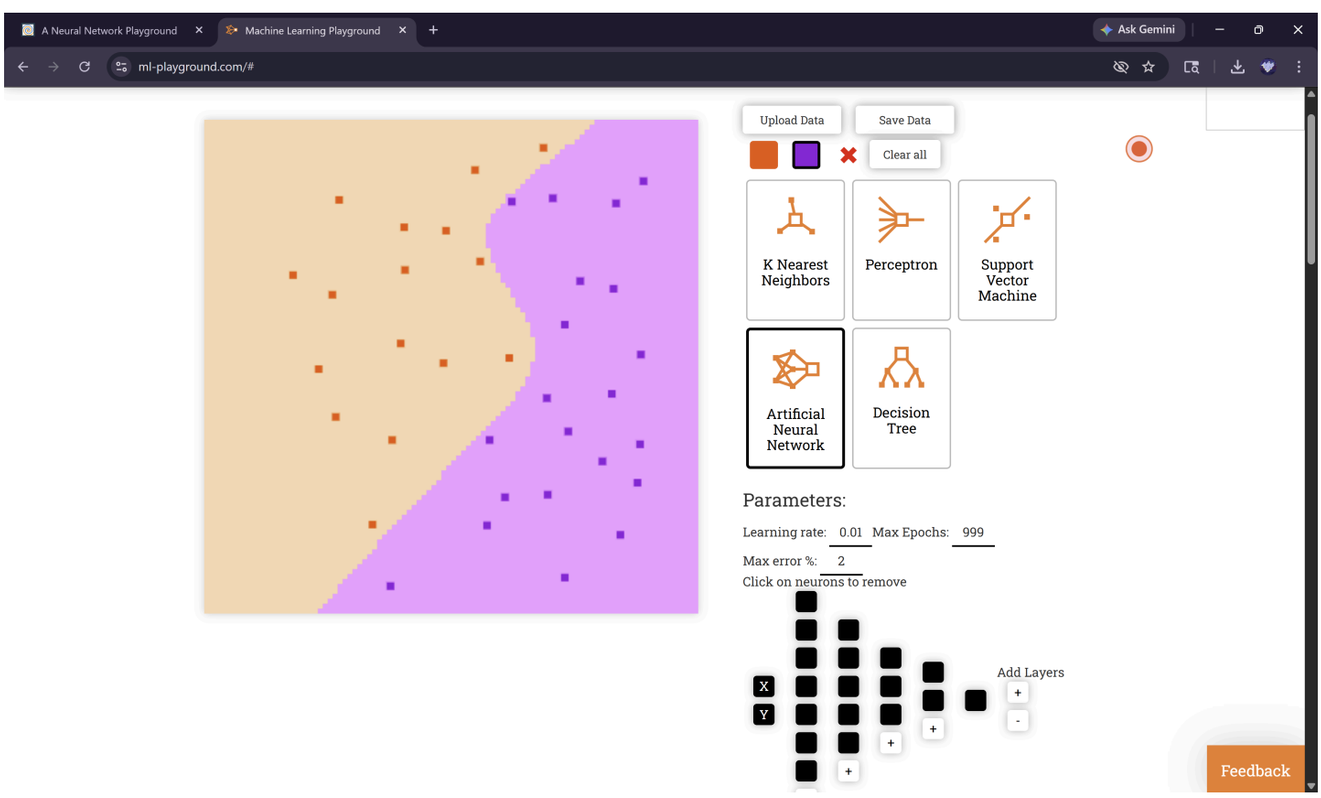
Double click here, and the text form will open for edition. Paste the screenshots below each line:

Task 1.3 (1) Underfit region:



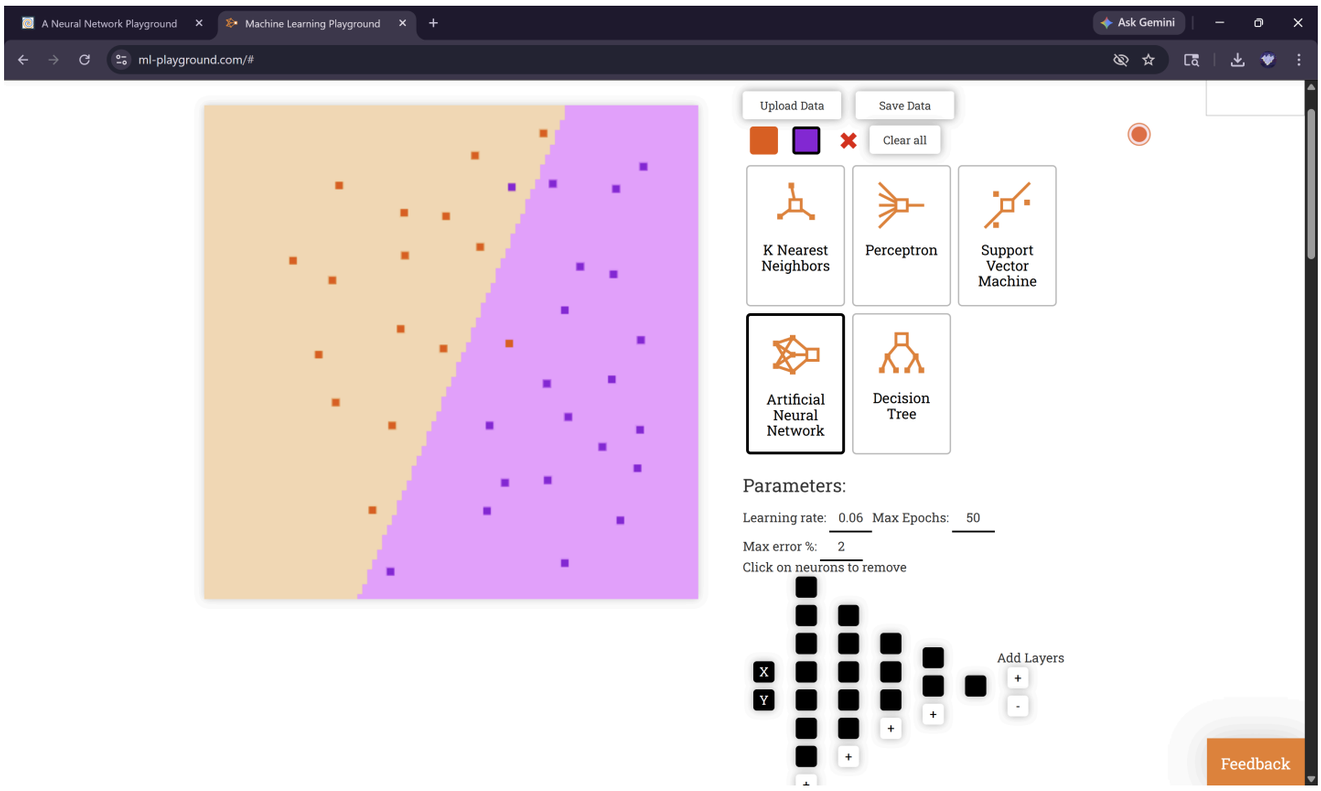
This model is underfitting because the decision boundary is too simple (almost a straight line) and does not capture the true pattern of the data. Many points are misclassified, especially near the boundary.

Task 1.3 (2) Overfit region:



This model is overfitting because the boundary is overly complex and wavy. It tries to perfectly classify all training points, including noise, which leads to poor generalization.

Task 1.3 (3)Fit region:



This model is well-fitted because the boundary appropriately separates the two classes without being too simple or too complex. It captures the general trend of the data.

Task 1.4

Looking at your visual graphs, what evidence do you have to support your belief whether the model is fit/underfit/overfit? Why does the model exhibit those fit-ness characteristics?

In the underfit model, the boundary is very simple (almost linear) and fails to correctly separate many data points from different classes. You can see several orange and purple points lying on the wrong side of the boundary. This indicates the model has not learned the underlying pattern of the data. The reason for this behavior is that the model is too simple (e.g., fewer layers, fewer neurons, or insufficient training), so it cannot capture the complexity of the dataset.

In the overfit model, the boundary is highly irregular and bends around individual data points. It closely follows even small fluctuations and noise in the dataset, often wrapping tightly around points. While it may classify training data almost perfectly, this complexity suggests poor generalization to new data. This happens because the model is too complex (e.g., too many neurons/layers or too many training epochs), causing it to memorize rather than learn general patterns.

In the fit model, the boundary is smooth and reasonably curved, separating the two classes with minimal misclassification. It captures the overall structure of the data without reacting to noise. This indicates a good balance between model complexity and training, allowing it to generalize well.

Interpreting Air-tight Vacuum and Air-leak Vacuum Data

Proceed to the next Colab Notebook to work in building a model on anomaly detection in a vacuum pump example in Prelab 8.2.

Please continue to Prelab 8.2 here.